Co-packaged Optics Technology Evaluation for Hyperscale Data Center Fabric Switches

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Abstract *We present a large-scale evaluation of the 51.2T CPO switch system, sharing statistical data on power consumption, optical performance characterization, and link reliability from over one million 400Gbps-port device hours of operation under stressed temperature condition. ©2025 The Author(s)*

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Introduction

The explosive growth in the size of large language models (LLMs) and the artificial intelligence (AI) training clusters required to enable them has led to data center power availability becoming a key consideration in infrastructure build-out. The simultaneous growth in graphics processing unit (GPU) node count and bandwidth, drives an increase in power needed for the scale-out and scale-up networks, and as a result has become a significant contributor to the data center power consumption. New technology solutions which improve the network infrastructure power efficiency are urgently needed to support future large cluster deployments.

The state-of-the-art solutions for data center network connectivity for use cases spanning tens of meters in-row links to tens of kilometers inter-building links are re-timed pluggable optical transceivers operating at 200 Gbps 4-level pulse amplitude modulation (PAM4) per lane. A variety of solutions across the pluggable optical module ecosystem are being explored to improve the power efficiency of these modules, including digital signal processors (DSPs) manufactured in advanced complementary metal-oxide-semiconductor (CMOS) process nodes, linear pluggable optics (LPO) which remove the DSP and its power consumption completely and even higher channel count optical transceivers which save power through higher levels of integration [1-3].

One key solution to the challenge of optical networking power consumption is co-packaged optics (CPO), which integrates the optical IO technology onto the switch package to both reduce the impact of electrical channel loss and to support higher channel count integration [4-6]. One significant challenge of this proposed solution, which has yet to be addressed, is the implication of the reliability, availability and serviceability (RAS) of the architecture to the network performance. To better understand the feasibility of CPO technology for high-volume deployments, we have developed a large-scale hardware and monitoring software infrastructure that aims to investigate CPO RAS.

In this paper, we provide an overview of Bailly 51.2Tbps CPO switch system design from Broadcom. We compare power consumption of the optics in the CPO switch to the measured power from pluggable optical modules and share our measurement results on key transmitter and receiver optical performance metrics. Finally, we present the results from our large-scale system-level reliability testing with over 1 million device hours of collected data on CPO switches.

System overview

For the CPO technology evaluation, we use Bailly 51.2Tbps CPO switch with eight 6.4Tbps silicon photonics optical engines (OEs) integrated onto the Tomahawk5 switch substrate. The switch system provides 128 ports of 400Gbps FR4 connectivity over standard single-mode fiber and enables 51.2 Tbps switching capacity [7].

The Bailly CPO package is attached to the switch main board (SMB), where the SMB provides the power and control functions to the Baily module. The switch system is integrated into a 4-rack unit (RU) platform with 128 LC duplex fiber connectors on the front panel. Additionally, the system is designed to work with field-replaceable pluggable laser sources (PLSs) accessible through the front panel. Each PLS supports eight 400Gbps ports.

The platform is air-cooled through 8 fans located at the back of the box. The rotation speed of the fans is automatically adjusted based on temperature through software control.

Power consumption

CPO technology has the advantage of reduced power consumption compared to legacy pluggable optical modules due to the tighter integration and reduced electrical channel loss between the application-specific integrated circuit (ASIC) and optics host interface.

Fig. 1(a) shows the measured optics power consumption per 800 Gbps bandwidth vs. OE die temperature normalized to the lowest OE temperature on 16 OEs and 32 PLSs from two CPO units at room temperature. We varied the OE die

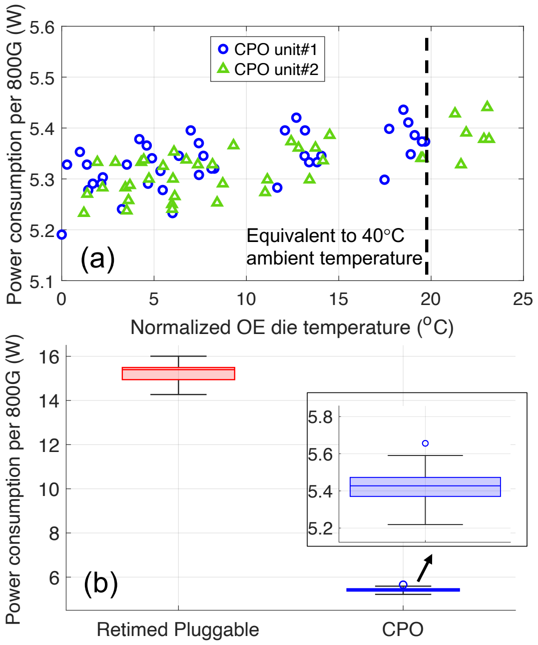

Fig. 1: a) Optics power consumption at different OE die temperatures. b) Power consumption comparison of 2x400Gbps FR4 pluggable and CPO. Inset: magnified CPO power.

temperature by manually changing the rotation speed of the fans. The increase in power consumption at higher die temperatures is negligible even beyond the stressed condition of 40°C ambient temperature.

In Fig. 1(b), we compare statistical data on 120 OEs and 240 PLSs from 15 CPO switch systems measured at 40°C ambient temperature, and 48 retimed 2x400Gbps FR4 pluggable modules from four vendors at 70°C module case temperature, corresponding to the worst-case operation temperature in a Minipack3 system [8]. Our measurements show a 65% optics power reduction from CPO compared to retimed pluggable optics, resulting in over 500W savings when compared to a fully populated Minipack3 system.

Optical performance

With the tight integration of many optical and electrical components together and co-packaging of the optical modules with the ASIC, the performance of the optical transmitters and receivers must be carefully studied to ensure a reliable data transfer between two end points. Additionally, interoperability with the existing pluggable optical modules must be verified through transmitter and receiver compliance tests.

We measured the key transmitter and receiver performance metrics on all ports of a CPO switch system. While the unit was at room temperature, we used the fan rotation speed setting to emulate the high temperature (HT) and nominal temperature (NT) operation condition at the system level.

Fig. 2(a), 2(b) and 2(c) respectively show the extinction ratio (ER), transmitter and dispersion eye closure quaternary (TDECQ), and optical

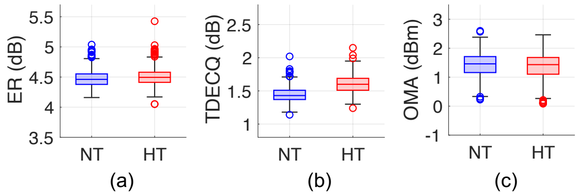

Fig. 2: Transmitter optical performance data. a) ER, b) TDECQ, and c) OMA distribution.

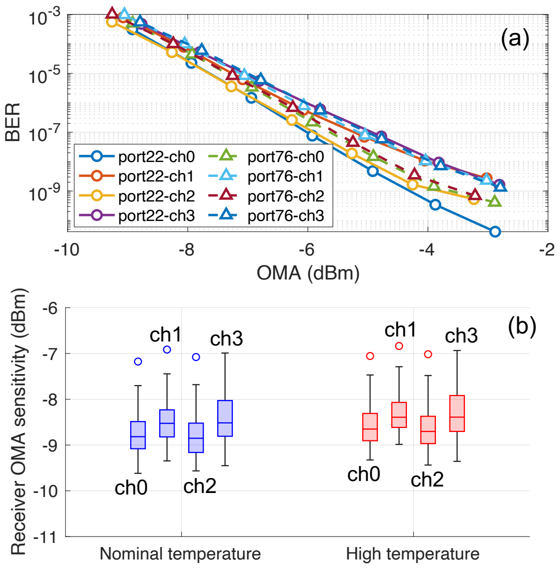

Fig. 3: a) BER vs. receiver OMA on two randomly selected ports. b) Receiver OMA sensitivity distribution on all ports.

modulation amplitude (OMA) measured on all 512 transmitter channels of the CPO unit in 100Gbps/lane mode. The measurements show margin with respect to the 400GBASE-FR4 standard specification [9].

Fig. 3(a) shows the bit error ratio (BER) vs. received OMA on all lanes of two randomly selected ports of the CPO unit. For this set of measurements, we used a 400Gbps FR4 pluggable optical module with TDECQ of approximately 2 dB and ER of around 5 dB across all 4 lanes as a reference transmitter. Fig. 3(b) shows the statistical distribution of receiver OMA sensitivity at a BER of 2.4×10^{-4} on all 512 lanes of the CPO unit. While not shown here, the BER floor was less than 5×10^{-8} across all 512 channels.

The performance consistency and margin to specifications play an important role in robustness of CPO manufacturing considering the large number of channels integrated together and lead to more reliable links during operation.

Link reliability and performance

One of the key challenges of deploying CPO technology at a large volume in hyperscale data centers is the serviceability and large blast radius. In systems using pluggable optical modules, any optics-related failure would be addressed simply by replacing the pluggable module. With the co-packaging of optics and ASIC, optics is not a field replaceable component anymore. For example, in the case of Bailly CPO, any failure in the OE would lead to a swap of the whole switch system chassis, which in turn has implications on link interruption rate, service time and cost.

CPO systems designed with field-replaceable PLSs not only eliminate this problem for one of the major failing components in optical modules but also mitigate quality and supply chain risks through multi-sourcing and improve laser reliability due to the reduced laser operation temperature. The laser junction temperature is typically less than 50°C in the worst-case operation condition of CPO systems, while in contrast pluggable transceivers typically rely on cooling the lasers via thermoelectric coolers (TECs) or use uncooled lasers with junction temperatures at around 80°C.

For CPO technology to become a viable solution in data center deployments at a large scale, the failure rate of OEs needs to be at a similar or lower level than other components integrated in the system. One may argue that with the integration of optics into the switch substrate and reduction of the number of interfaces compared to a pluggable optical module, the failure rate of the optics could be significantly reduced, making CPO a feasible technology for deployment from the reliability point of view. To test this hypothesis and to evaluate the link reliability and performance of the CPO technology, we have set up a large-scale CPO switch test infrastructure, where we have been collecting key reliability and performance metrics every 5 minutes at stressed ambient temperature condition of 40°C. The CPO units are operating in 128x400Gbps mode with optical self-loopback fibers on all the ports.

Fig. 4(a), 4(b) and 4(c) respectively show per channel transmit laser bias current, transmit and received power, and signal-to-noise ratio (SNR) data from one port of a randomly selected CPO unit over 15 days. The demonstrated robustness in link parameters over time is required to minimize the link failures at a large scale.

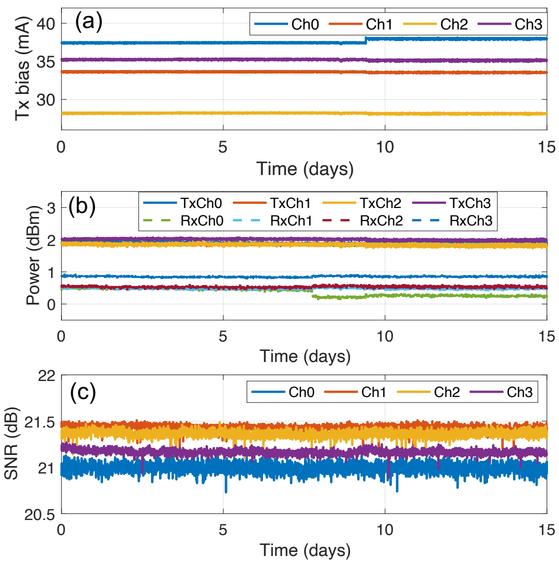

Fig. 4: a) Transmitter (Tx) laser bias current, b) Tx and receiver (Rx) power, and c) SNR over 15 days of operation

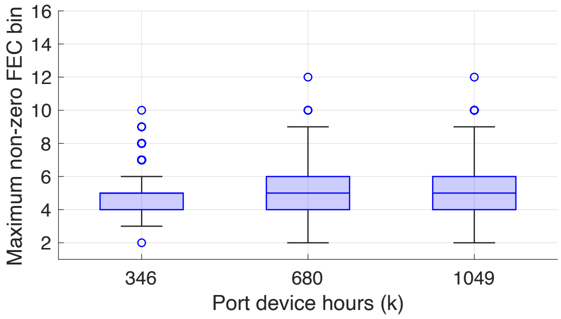

Fig. 5: Maximum non-zero FEC bin vs. 400Gbps-port device hours of operation at 40°C ambient temperature.

Fig. 5 shows the distribution of the maximum non-zero KP4 forward error correction (FEC) bin observed across 128 ports of all CPO units vs. 400G-port device hours of operation. To the best of our knowledge, this is the highest device hours reported for CPO technology operation at the system level. Over the period of this experiment, each unit ran continuously without interruption or clearing the FEC counters and we did not see any failures or uncorrectable codewords (UCWs) in the links. The maximum non-zero FEC bin is smaller than 7 on 75% of the ports after 1.05 million device hours of operation. One instance of FEC bin > 10 has been seen in this period. We have been able to correlate the performance of this port to the manufacturing line test data and confirmed this port has not degraded.

The demonstrated lower bound for mean time between failures (MTBF) of optical links can readily support a 24k GPU AI cluster with >90% training efficiency without interconnect failures being the bottleneck [10,11].

Conclusions

As the switch bandwidth increases from one generation to the next, the IO density, power and cooling of switching platforms pose further challenges. CPO technology can be a key enabler offering significant power, bandwidth and latency advantages for hyperscale data centers. We shared a large-scale evaluation of Bailly 51.2Tbps CPO switch power consumption, optical performance and link reliability. The CPO technology demonstrated a power saving of 65% compared to retimed pluggable optical modules in 100G/lane generation and supports interoperability through compliance to standard optical performance metrics. Additionally, our measurements from the long-term reliability test infrastructure show no UCWs for over 1 million 400Gbps-port device hours of continuous operation.

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